

# Research Methods in IT

## Lecturer 101: Foundations for Academic and Applied Inquiry

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### MSIT LECTURE HANDOUT

Course: Topics in IT    Program: MSIT

Problem Identification • Research Design • Ethics • Analysis • Reporting

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# 1 Why Research Methods Matter in IT

## 📖 Research Methods in IT

Research methods in IT are the structured principles and procedures used to identify a problem, collect trustworthy evidence, analyze findings, and produce defensible conclusions that can inform technology practice, policy, or innovation.

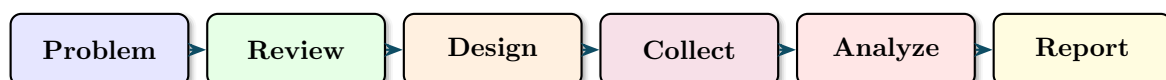
## 💡 Lecturer 101 View

At introductory level, research methods are less about advanced statistics and more about disciplined thinking. A strong IT study begins with a clear problem, a justified method, ethical evidence collection, and conclusions that actually answer the research question.

## 1.1 Why IT Professionals Need Research Skills

- To evaluate whether a new tool, system, or framework actually works.
- To justify IT investments using evidence rather than opinion.
- To investigate user problems, security issues, or process inefficiencies.
- To contribute to academic, industry, and organizational knowledge.
- To communicate findings in a professional and reproducible way.

## 1.2 Research as a Decision Process



| Stage                  | Key Question                                                 |
|------------------------|--------------------------------------------------------------|
| Problem identification | What issue in IT needs investigation?                        |
| Literature review      | What is already known and where is the gap?                  |
| Research design        | Which method best answers the question?                      |
| Data collection        | What evidence will be gathered and from whom?                |
| Analysis               | How will findings be interpreted?                            |
| Reporting              | What conclusions, implications, and limits should be stated? |

# 2 From Topic to Research Problem

## 2.1 Topic, Problem, and Gap

### 📖 Research Problem

A research problem is a specific and significant issue that requires systematic investigation. It is narrower than a broad topic and clearer than a general interest area.

### From Broad Topic to Problem

Broad topic: *Cybersecurity awareness in universities.*

Better research problem: *Despite repeated awareness seminars, phishing click rates among graduate students remain high, and the factors influencing unsafe email behavior are not clearly understood.*

## 2.2 Characteristics of a Good Research Problem

- Clear and focused.
- Relevant to IT practice or policy.
- Researchable using available time and resources.
- Supported by observable evidence.
- Valuable enough to justify the study.

## 2.3 Sources of Research Problems

- Persistent system failures or usability complaints.
- Security incidents or policy non-compliance.
- Low adoption of digital platforms.
- Gaps in prior studies.
- Contradictory results in the literature.

### Common Beginner Mistake

Many students choose a technology instead of a problem. “Blockchain” is not a research problem. “Barriers to blockchain adoption for academic credential verification in private universities” is closer to a real research problem.

## 3 Reviewing Related Literature

### 3.1 What a Literature Review Does

#### Purpose of the Review

The literature review shows what is already known, what methods others used, where findings disagree, and what gap your study intends to address.

- Defines concepts and variables.
- Identifies theories or models.
- Locates research gaps.
- Helps justify the chosen method.
- Prevents duplication of weak or outdated work.

### 3.2 What to Read

- Peer-reviewed journal articles.
- Conference papers in computing and IT.
- Standards, policies, and technical reports.
- Theses and dissertations when relevant.
- Credible institutional or industry publications.

### 3.3 How to Organize the Review

| Approach      | What It Looks Like                                                        |
|---------------|---------------------------------------------------------------------------|
| Thematic      | Group studies by themes such as usability, adoption, risk, or performance |
| Method-based  | Group by survey, experiment, case study, interview, or mixed method       |
| Chronological | Show how the field developed over time                                    |
| Model-based   | Compare studies using the same theory or framework                        |

#### ✎ Literature Matrix Idea

A simple literature matrix may include: author, year, context, method, sample, key findings, and gap. This helps the researcher avoid reading articles in isolation.

## 4 Research Questions, Objectives, and Hypotheses

### 4.1 Research Questions

Research questions guide the entire study. Good questions are specific, answerable, and aligned with the problem statement.

#### ✎ Sample Research Questions

Study topic: learning management system adoption among graduate students.

Possible questions:

1. What factors influence graduate students' use of the learning management system?
2. Is perceived ease of use associated with frequency of platform usage?
3. What challenges do students report during remote access?

### 4.2 Research Objectives

Objectives tell what the study intends to accomplish.

- **General objective:** broad purpose of the study.
- **Specific objectives:** measurable sub-goals that support the general objective.

### 4.3 Hypotheses

Hypotheses are testable statements, usually used in quantitative studies.

#### Hypothesis

A hypothesis is a tentative claim about the relationship between variables, stated in a form that can be supported or rejected through evidence.

#### Hypothesis Example

$H_0$  : Perceived ease of use has no significant relationship with system usage frequency.

$H_1$  : Perceived ease of use has a significant relationship with system usage frequency.

## 5 Research Designs Common in IT

### 5.1 Quantitative Research

Quantitative research focuses on numerical data, measurement, and statistical analysis.

- Surveys with scaled responses.
- Performance comparisons.
- Experiments and quasi-experiments.
- Log analysis and metrics evaluation.

### 5.2 Qualitative Research

Qualitative research focuses on meaning, experience, process, and context.

- Interviews with users or administrators.
- Focus group discussions.
- Observations of system use.
- Document or policy analysis.

### 5.3 Mixed Methods Research

Mixed methods combines numerical and narrative evidence in one study.

#### Why Mixed Methods Fits IT

Many IT problems have both technical and human dimensions. A survey may reveal how often a problem occurs, while interviews explain why it occurs.

### 5.4 Common Designs at a Glance

| Design | Best For | Typical IT Example |
|--------|----------|--------------------|
|--------|----------|--------------------|

|               |                                   |                                            |
|---------------|-----------------------------------|--------------------------------------------|
| Survey        | Describing attitudes or practices | User satisfaction with campus Wi-Fi        |
| Experiment    | Testing cause-effect claims       | Comparing two authentication interfaces    |
| Case study    | Deep contextual investigation     | Digital transformation in one university   |
| Correlational | Examining relationships           | Security awareness and password behavior   |
| Mixed methods | Combining breadth and depth       | LMS adoption rates plus student interviews |

## 6 Sampling and Participants

### 6.1 Population and Sampling Frame

The population is the full group of interest. The sampling frame is the practical list or source from which participants are selected.

#### ✎ Sampling Frame Example

If the population is all MSIT students in a university, the sampling frame might be the official graduate student enrollment list for the current semester.

### 6.2 Probability and Non-Probability Sampling

- **Probability sampling:** participants are selected with known chances, such as simple random sampling.
- **Non-probability sampling:** participants are selected through convenience, purposive choice, or availability.

### 6.3 Examples of Sampling Techniques

| Technique              | Use Case                                                            |
|------------------------|---------------------------------------------------------------------|
| Simple random sampling | When a complete list of participants exists                         |
| Stratified sampling    | When subgroups such as year level or role should be represented     |
| Convenience sampling   | When quick access matters more than strong generalization           |
| Purposive sampling     | When specialized participants are needed, such as IT administrators |
| Snowball sampling      | When participants help identify others with similar expertise       |

## 6.4 Simple Response Rate Calculation

### ⌕ Response Rate Example

Suppose 150 questionnaires were distributed and 123 were returned.

$$\text{Response rate} = \frac{123}{150} \times 100 = 82\%$$

This is a useful descriptive statistic for reporting data collection quality.

## 7 Instruments, Validity, and Reliability

### 7.1 Research Instruments

An instrument is the tool used to collect data.

- Questionnaire or survey form.
- Interview guide.
- Observation checklist.
- Test script, rubric, or performance log template.

### 7.2 Validity

Validity asks whether the instrument measures what it is supposed to measure.

- **Content validity:** items cover the concept adequately.
- **Construct validity:** items align with the underlying concept.
- **Face validity:** the instrument appears reasonable to experts or respondents.

### 7.3 Reliability

Reliability asks whether the instrument produces consistent results.

### 💡 Practical Meaning

If a usability scale gives widely different results for the same situation without good reason, the instrument may have poor reliability.

### 7.4 Likert-Scale Mean Example

### ⌕ Mean Rating Example

Suppose five respondents rate system usability on a 1–5 scale as follows:

4, 5, 3, 4, 4

The mean rating is

$$\bar{x} = \frac{4 + 5 + 3 + 4 + 4}{5} = \frac{20}{5} = 4.0$$

This suggests generally positive usability perception.

## 8 Data Collection and Research Ethics

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### 8.1 Common Data Collection Methods

- Online or printed surveys.
- Semi-structured interviews.
- Focus group discussions.
- System usage logs and audit trails.
- Controlled task performance observations.

### 8.2 Ethical Principles

- Informed consent.
- Voluntary participation.
- Privacy and confidentiality.
- Data security and secure storage.
- Honesty in reporting and citation.

#### Ethics in IT Research

IT research often involves digital traces, logs, or identifiable online behavior. A dataset that appears technical may still contain personal or sensitive information and must be handled carefully.

### 8.3 Examples of Ethical Risks

- Collecting login logs without proper permission.
- Recording interviews without explicit consent.
- Reporting findings in a way that identifies participants.
- Reusing a questionnaire without proper acknowledgment.

## 9 Data Analysis Basics for Lecturer 101

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### 9.1 Quantitative Analysis

At introductory level, quantitative analysis often starts with descriptive statistics:

- Frequency and percentage.
- Mean and standard deviation.
- Ranking of factors.
- Correlation or basic hypothesis testing when appropriate.

## 9.2 Qualitative Analysis

Qualitative analysis often involves coding and theme development.

1. Read the responses carefully.
2. Identify recurring ideas or phrases.
3. Group them into codes.
4. Merge related codes into themes.
5. Interpret the themes in relation to the research questions.

## 9.3 Simple Frequency Example

### Percentage Example

Suppose 48 out of 60 respondents say they use two-factor authentication.

$$\text{Percentage} = \frac{48}{60} \times 100 = 80\%$$

This indicates that most respondents report using two-factor authentication.

## 9.4 Interpreting Results

### Interpretation Rule

Do not stop at the numbers. Ask what they mean for IT practice. A high mean on usability should connect to design quality, support, adoption, or user productivity.

# 10 Writing the Research Proposal or Report

## 10.1 Typical Sections

| Section                        | Main Purpose                                            |
|--------------------------------|---------------------------------------------------------|
| Introduction                   | Present the background and research problem             |
| Review of literature           | Show what is known and where the gap lies               |
| Methodology                    | Explain design, participants, instruments, and analysis |
| Results                        | Present findings clearly and accurately                 |
| Discussion                     | Explain the meaning and implications of findings        |
| Conclusion and recommendations | Summarize the study and suggest next steps              |

## 10.2 Good Academic Writing Practices

- Write clearly and avoid unsupported claims.
- Align every section with the research questions.
- Use citations consistently.

- Report limitations honestly.
- Distinguish between findings, interpretation, and recommendation.

#### Weak Conclusion Pattern

Avoid generic endings such as “the system should be improved.” A stronger conclusion specifies which part should be improved, based on which evidence, and why it matters.

## 11 Mini Research Scenarios in IT

### 11.1 Scenario 1: LMS Adoption

Problem: Graduate students underuse the LMS despite institutional support.

Possible method: survey plus follow-up interviews.

Possible variables: perceived usefulness, ease of use, training quality, access reliability.

### 11.2 Scenario 2: Phishing Awareness

Problem: Staff continue clicking suspicious links after awareness campaigns.

Possible method: correlational survey and simulated phishing exercise.

Possible outputs: percentage of users who recognize phishing patterns, mean awareness score, thematic reasons for risky behavior.

### 11.3 Scenario 3: Helpdesk Service Quality

Problem: Users perceive IT support as slow and inconsistent.

Possible method: case study using helpdesk logs, survey ratings, and interviews.

#### How to Think Like a Researcher

For each scenario, ask four things:

1. What is the exact problem?
2. What evidence is needed?
3. Which method fits that evidence?
4. How will the findings help practice or policy?

## 12 Quick Reference Summary

| Concept           | Key Idea                                           |
|-------------------|----------------------------------------------------|
| Research problem  | Specific issue requiring systematic investigation  |
| Research gap      | What prior work has not adequately answered        |
| Research question | Direct question the study must answer              |
| Objective         | What the study intends to accomplish               |
| Hypothesis        | Testable statement about variables                 |
| Research design   | Overall plan for collecting and analyzing evidence |

|             |                                                 |
|-------------|-------------------------------------------------|
| Sampling    | Selecting participants from a population        |
| Validity    | Accuracy of what an instrument measures         |
| Reliability | Consistency of measurement                      |
| Ethics      | Protection of participants and honest reporting |
| Analysis    | Turning raw evidence into meaningful findings   |

## 13 Key Takeaways and Review Questions

### 13.1 Key Takeaways

- Good IT research begins with a real problem, not just a trendy technology.
- The literature review justifies the gap and sharpens the study.
- Research questions, objectives, and hypotheses must align.
- Method choice depends on the nature of the evidence needed.
- Reliable instruments and ethical practice are non-negotiable.
- Results should be interpreted for IT practice, not only reported as numbers or quotes.

### 13.2 Review Questions

1. What is the difference between a broad topic and a research problem?
2. Why is a literature review necessary before collecting data?
3. When is a hypothesis useful in an IT study?
4. How do quantitative and qualitative designs differ?
5. Why are validity and reliability both important?
6. What ethical issues commonly arise in IT research?
7. What makes a conclusion academically strong?

#### Short Reflection Task

Choose one IT issue from your workplace, institution, or community. Write one problem statement, two research questions, one suitable design, and one ethical concern you would need to manage.

### End of Handout

Research Methods in IT — MSIT Program